Traffic Flow and Congestion Analysis

Program explanation and documentation

1. Aim

To develop a Python program that analyzes traffic volume on multiple roads over a number of time intervals, determines congestion levels, checks whether road capacity is exceeded, and finds the highest recorded traffic count.

2. Description

The program first accepts the number of roads, the number of time intervals, and the road capacity threshold. For each road, it reads the road name and vehicle count for every interval. It calculates the total and average traffic, classifies congestion as Low, Moderate, or High, and reports whether any interval exceeded the capacity. It also tracks the highest single-interval vehicle count across all roads.

3. Input

Input	Purpose
Number of roads	Specifies how many roads will be analyzed.
Number of time intervals	Specifies how many vehicle-count readings are collected per road.
Road capacity threshold	Limit used for interval capacity checks and congestion classification.
Road name	Identifies each road.
Vehicle count	Number of vehicles recorded during each interval.

4. Processing Logic / Algorithm

1. Start the program.

2. Read the number of roads, number of time intervals, and road capacity threshold.

3. Initialize peak traffic to 0, peak road to an empty string, and peak interval to 0.

4. For each road, read its name and initialize total vehicles to 0 and the exceeded flag to False.

5. For each time interval, read the vehicle count and add it to the road's total.

6. If the current count is greater than the peak traffic value, update peak traffic, peak road, and peak interval.

7. If the current count is greater than the capacity threshold, set the exceeded flag to True.

8. After all intervals for the road, calculate average traffic by dividing total vehicles by the number of intervals.

9. Classify congestion: Low if average is below half the capacity; Moderate if average is up to and including capacity; otherwise High.

10. Display the road name, average traffic, congestion level, and capacity status.

11. After all roads are processed, display the road, interval, and vehicle count for peak traffic.

12. Stop.

5. Output

For each road, the program displays its average vehicle count, congestion classification, and whether any interval exceeded capacity. At the end, it displays the road name, interval number, and vehicle count corresponding to the highest single-interval traffic count.

6. Congestion Classification

Condition based on average	Congestion level
Average < 50% of capacity	Low
50% of capacity ≤ Average ≤ Capacity	Moderate
Average > Capacity	High

7. Important Notes

• Capacity is considered exceeded only when an interval count is strictly greater than the capacity threshold.
• Congestion classification is based on the road's average traffic, not on whether an individual interval exceeded capacity.
• Peak traffic is the largest single interval count across all roads. If the maximum occurs more than once, the program retains the first occurrence because it updates only when a strictly greater count is found.
• The program assumes that the number of intervals is greater than zero.

8. Conclusion

The program summarizes traffic conditions across multiple roads, identifies possible capacity issues, categorizes average congestion, and reports the peak traffic period. It can support basic traffic monitoring and comparison.

Note: This is a basic analysis using user-entered vehicle counts and a fixed capacity threshold; it does not use live traffic data.